# Configuration-level split aliasing in object-detection benchmarks: what the reported split is selected on, and how much of the selection survives an independent split


## Abstract

Object-detection papers report a number from a split they call held out. We audit that premise, in three results.

**1. The convention is broken in our own corpora. 6 of 14** corpora give the reported number no independent held-out split: the checkpoint was selected there, **even though** training and evaluation do not overlap. What is broken is **checkpoint selection**, not contamination — which turns an ethical question into a **measurement** one.

**2. The selection premium only partly survives.** The premium is positive in every escalated cell; what changes sign by corpus is the **paired Δgap** — the arms' relative difference on `val` versus on `test` — and not the selection term itself. The **mean direction is metric-robust while one cell's significance and sign unanimity are not** (`smoke2sf`, p = 0.54). A declared, **descriptive** 13-cell scan leaves **3** cells after BH, so the direction claim rests on the four escalated cells. **On a disjoint split, realization is partial**: the **claim-level interval includes zero**, and the arm-cell and run-weighted point estimates are **descriptive**, given with intervals only (Table 4). No power analysis was registered, so the scan's nine `n` = 3 cells stay descriptive.

**3. The printed count rests on one rater.** The layer the literature actually reports is printed as **12–14 of 19**; re-verifying every row first-hand and re-coding blind with **five** current-generation models recovers **19 of 30** corrected cells, eleven unanimously, while **26 of 76** stay undecidable and **11 of the 30 are not recovered** (§6.1, S12).



**Keywords:** evaluation validity; dataset splits; checkpoint selection; object detection; benchmark audit; reproducibility


## 1. Introduction

A detection paper reports mAP50-95 on a split it calls held out, implying a number that estimates performance on unseen data. The inference needs a premise the paper rarely states: the checkpoint was not chosen on that same split. Binding `val:` and `test:` to one directory breaks it silently — the pipeline evaluates there, keeps the best-scoring checkpoint and reports its score: a

**maximum**, not a sample. Surveys of the detection literature [1] trace how that number became the field's unit of comparison; this paper audits the reporting convention producing it.

That premise is audited in three places, in increasing distance from our own work: **our own corpora**, where we count how many of our corpora alias the two keys (19 YAML files, counting by file); **19 public benchmarks**, audited row by row for which split the reported number comes from; and **20 published third-party training logs**, where we measure the premium directly, in data we did not produce.

Three further things we have not seen done here. We **measure the premium's fate** under a protocol separating the selection set from the reported set (`best` and `last` checkpoints and both splits exist, so the full 2 × 2 grid is computable): how much of the `val`-selected premium is realized on an independent split — the measurement this paper contributes and §3.1 formalises. The protocol is the companion's and is **cited, not restated**. And we **pre-register** a generalization hypothesis, then report that it misses its criterion — a registration **also reported in the companion**.

### 1.1 Why the unit matters

Ask "how many of these benchmarks have a split problem?" and, unless a unit is declared, every answer is defensible and they disagree by a factor of six: 2/19 (**structural**: one path bound to both keys), 10/19 (**`release` unit**: no independent held-out test), 13/19 (**`protocol` unit**: local independent testing gated), **12–14 of the 19** (**`reported` unit** — **12** the single rater's printed coding, **14** that rater's rows after a **non-blind recoding** moved two; **five blind coders agree at κ 0.824** and, once the record is re-verified first-hand, independently reach **19 of the 30** corrected cells, eleven unanimously (**S12**); the reported number's split is non-independent). The readings above are all defensible, and §6.1's four counting units (`release`, `protocol`, `yolo_dist`, `reported`) are the **columns** they come from — including `yolo_dist`, omitted above, at **4/19**. Declaring the unit is therefore a methodological premise, not a caveat (§3, §6).

### 1.2 What we do not claim

We do not claim that aliasing is data contamination: the image-level intersection between the training and held-out evaluation images is **zero** (SHWD excepted, where `train` and `val` are one directory). We do not claim a single bias

direction: our paired analysis finds **two corpora where `val` understates and two where it overstates**, and **nine cells are not testable at n = 3** (§5.1, §7.1). Nor do we claim the convention is framework-general (§6.5) or that the audited benchmarks are wrong: they report what their configuration computes.

**Positioning (one sentence).** We claim neither to be first to audit split problems in detection benchmarks nor to audit image-level leakage: both are taken. We occupy a different cell — **configuration-level aliasing, a direction that varies by corpus, and the requirement that a count declare its unit.** Every number points to a local artifact; every claim narrowed after external checking is recorded as narrowed (§5.4).

**Scope.** All quantitative claims about selection behaviour come from **YOLO-lineage pipelines**, the only family among the six frameworks we checked (five pinned repositories plus the Ultralytics release) selecting the checkpoint by the validation metric by default (§6.5) — stated at the claim, not only in a limitations section. **The registered multi-target replication is reported in the companion paper, and its first report, criterion and verdict stay there** (§9): this paper cites it for readability and **claims none of its results**.

## 2. Contributions and related work

**Where this paper sits.** Reporting practice has itself been audited: leakage and near-duplicate contamination [3], [4], [11]; split naming and the choice of standard splits [8], [9]; experiment- reporting standards [7]; checkpoint-selection criteria [6]; and budget-dependent rankings [2]. Each audits

**data, reporting, or selection**; **none audits a release's configuration binding** — the object of §3–§6.

### 2.1 The closest work, and where this paper sits (detailed)

> The detailed comparison — the nine prior lines of work, what each measures, and the axis on which

> this paper differs — is **Table S45 in the Supplementary material**; it is not printed here because

> the journal counts text inside the 20–35 page limit and the comparison is background, not a claim.

> **§2.2 (Contributions) and §2.3 (what we do not claim) are unaffected.**

## 2.2 Contributions

**What this paper reports first** (the boundary to the companion is in §2.3): the configuration-level audit of the `val`/`test` convention and its selection premium (§4–5); and the **measured** realization of that premium on a disjoint split with the reporting conventions it depends on (§8.6–8.7, §11, §3.1).

1. **We quantify the magnitude and direction of configuration-level aliasing.** The selection premium (best − last) is **positive in all four cells, the baseline arm gaining more** (**near-definitional; §5.1**; four paired 95 % CIs excluding zero, §5.1 Tier 1) — **consistent with the selection-bias literature** [5], not new in kind. **All four cells meet the stated-n criterion (n = 10–11)** and all four survive a Benjamini–Hochberg step over **these four** at q = 0.05 — the fourth, `dota15`, entered at n = 3 and was escalated to n = 10 under a criterion fixed in advance (**Table S28's note**) — so the convention systematically shifts **arm-to-arm comparisons**; its direction varies by corpus: three-way paired differencing gives 2 cells where `val` understates, 2 where it overstates and 9 not testable at n = 3 — the **no-systematic-direction** [10] reports, measured here on detection corpora.

**We do not claim a general understatement** (§5). It appears in third-party logs too (a searched 20-log sample; §5.5) — corroboration that the component exists elsewhere, **not independent evidence** and not a field-level rate.

*Note: image-level** leakage magnitude is quantified by the two 2026 studies; **configuration-level** aliasing magnitude has not been measured before.*

2. **We measure how far the selection premium transfers.** The measurement is the contribution; the protocol that makes it readable is the companion's and is **cited, not restated** here (§8.1). On the five arm-cells measured here the `val`-selected premium is realized on the disjoint split at

**23.3 %** with all five arm-cells at n = 10, and it is **not a constant**: the rate falls with the validation noise of the criterion used to select (§8.6–§8.7). The identity tying it to the winner's-curse share we **adopt** [13], not claim; the closest concurrent work [12, 15] reaches the same **direction** with a different quantity, and neither reports a **ratio of difference premiums** — `(T_best − T_final)/(V_max − V_final)`, on disjoint splits, in detection, on our own arm-cells, with measured `val` noise as moderator. A same-estimand ratio does exist elsewhere — in genomics, where it is written as a relative bias and

measured at 10–60 % (**Supplementary S12** carries the full citation). The replicate axis is **data order** (§8.4) — the **seed-budget discipline reproduced inside this setting** rather than a new one.

3. **We disclose our own defects, and the corrections that followed** rather than removing them: our analysis' first version had non-reproducible items — pod-dependent scripts, a citation to a file not holding the quoted numbers, no written endpoint convention. §10, not deleted.

### 2.3 What we explicitly do not claim (shared tooling)

The companion and this paper share project-internal process protocols — a multi-model review checklist and a local/cloud scripting discipline — and neither claims originality for them: **Supplementary S6**, *Shared process protocols*.

## 3. Units: the methodological premise

"Have a split problem" is a property of a **(benchmark, counting unit)** pair, not of a benchmark. **[14] varies the definition on one file; we vary who controls the split and measure a different quantity: the share of `val`-selected premium that survives a disjoint split.** We use four units, defined by **who controls the split**:

- **`release`** — the split the benchmark's **own published release** ships: the `train:` / `val:` / `test:` entries and paths in the distributed archive. *Example*: an archive whose `data.yaml` binds `val:` and `test:` to the **same directory** records "yes" here (§4).

- **`protocol`** — the split the benchmark's **official protocol** tells you to use, which need not be the released one: a protocol may carve its own validation set out of the released training split.

*Example*: an aliasing release whose documented protocol still requires a held-out test split records "no" here and "yes" above — the units answer different questions.

- **`yolo_dist`** — the split a **generic YOLO distribution package's** own configuration resolves to, i.e. what a practitioner taking the pipeline off the shelf gets. *Example*: a clean release whose community YAML points `val:` and `test:` at one directory records "yes" **only** here.

- **`reported`** — the split the number **the literature prints** came from (not the official leaderboard's own score), read from the release documentation and the

dataset's own paper. *Example*: two papers on one benchmark can differ here while agreeing under all three units above — hence the headline count uses this unit (§6.1).

**A count is meaningless without its unit**: the same nineteen benchmarks answer differently under each (Fig. 1).

**The proposition is not ours to claim.** That *the same data yield different defensible audit counts under different operational definitions* is stated independently for molecular benchmark auditing, in a 2026 preprint whose abstract closes "different defensible definitions return different answers from the same file" [14]. **The observation is the cited literature's; its instantiation in detection evaluation, and the four named units, are ours** — the units named by *who controls the split*, the 19-row evidence table, and the premium on **five arm-cells**. It is orthogonal to [15], which separates leakage by **type**; we separate the **counting subject**, so one type can still be counted under four units (§6.1).

> **Table S2** → Supplementary §S8.

**The `reported` unit is read from the release documentation and from the dataset's own paper**: we did not sample papers per benchmark, so no prevalence claim is made (§11). For each of 19 benchmarks we record all four units, an evidence class and a confidence level, and recompute every percentage from the row markers under a per-unit subtotal check **and one stated `n_a` rule** (**Supplementary S12**) — the full 19 × 4 table and the cross-check against the original audit are **Supplementary S1**, the derived counts §6.1 and its table.

**The rule itself is not new.** The same toolchain's own course states it normatively — *"Test set is touched once, at the end. Not for tuning."* and *"The splits must be independent on the dimension that matters at deployment."* [16] — so what this paper adds is not the rule but the **audit and the count** under it.

**The premise we rely on throughout:** a "yes" under any unit means *the reported number is not a sample from an independent split*. **What that does and does not license is stated where the counts are printed** (§6.1 and the note under its table).

### 3.1 Formalisation: what the four quantities are, and what they can and cannot identify

**Setup.** Training is a randomised map from (seed ω, recipe θ, checkpoint e) to weights, and evaluation a fixed functional on a labelled split; write V(e) and T(e)

for a checkpoint's `val` and `test` readings and ê for the `val` argmax. Sections 5–9 measure four quantities: the selection premium, Δgap, κ, the realization rate (**notation and formal statements: Supplementary S11**).

**Definition — the raw gap is not identifiable from the pair alone.** G = V(ê) − T(ê) = δ + o mixes a split-difficulty term with a selection-optimism term; the decomposition, **and the reading that a single-model endpoint pair is unable to separate its two terms, are [17]'s**, while its **strong form** — that *any* δ′ fits the observation equally well — is **stated here as a formalisation rather than attributed to that literature**. **The pairing assumption is what makes Δgap meaningful**, failing by construction under aliasing so that §9 cannot be read as a clean-protocol result; the identity holds value by value on all 13 scanned cells and both escalated cells at n = 10 (§5.6).

**Adopted identity, not a claim of this paper.** With V(e) = v(e) + ε_e and T(e) = v(e) + τ + η_e, **E[premium_test] = E[premium_val] − WC**, WC := E[ε_ê − ε_e_f] ≥ 0 — the argmax is over a **candidate set fixed in advance** and `e_f` is a **pre-specified member of it** (what the budget-fixed `last` gives) — so the realization rate is **1 − WC / E[premium_val] ≤ 1**, with equality only if selection is driven by signal alone [13] (**Supplementary S11** states which part of that the archive can and cannot settle) — the optimizer's curse in this paper's notation, the frame for §8.6–§8.7. **Two consequences are testable on the archive** — the per-run inequality, and monotonicity in the `val` noise (**Supplementary S11**). A third form, that the premium scales as σ√(2 ln E), was **tested and withdrawn** (**Supplementary S11**); contemporaneous work measures ≈**5**× on the same σ√(2 ln K) bound [18].

**Corollary (the premium is mostly shape, and the peak is the worst place to stand).** `premium_r = shape(E) + v_r` holds **exactly**, `shape(E)` being the mean curve's peak-to-final gap, so `shape` is a functional of the mean curve alone — **not** an independent check on σ — and `E[v_r] ≥ 0`. On our 40-run four-budget series **the monotone criterion is met on the clean re-run** while the σ√(2 ln E) form is **outside** its pre-declared band, so that scaling **stays withdrawn**; the pre-declared shape control passes on the mean it specifies but fails per arm, and we report both readings — the with-early-stopping series carrying the pre-declared verdict — so **we do not claim an `E` effect**. The checkpoint rule is a **reporting rule** (the top-5 mean beats the argmax on the

same 40 runs), with **three limits**: an empirical capture rate, a `val`-side cost the argmax does not carry, and a two-split, five-epoch-grid scope.

**What is ours, and what is not.** Of the statements that follow, the raw gap's definition [17], the impossibility result for endpoint-pair estimators [19] and the family-dependence procedure [20] are **machinery rather than results of ours**, and the identity is **adopted** from the selection-bias literature; **the corollary alone carries measurements of this paper**. Full statements and proofs are in **Supplementary S11**, and the detail displaced from here in **Supplementary S12**.

### 3.2 Six experiments, five judged criteria, on GPU machines, each with its criterion written down first

The propositions of §S11 are testable, so we ran them on rented GPUs with the **criterion fixed before the runs**; **Supplementary Table S37** gives, for each of the six experiments, the criterion it was judged against. Counts below are over **experiments**, not criteria.

**The two negative outcomes and the redone design are stated as such in Supplementary Table S37** (G1 is void; **two of the five judged experiments are negative — G1′ and G2**; the values behind every verdict are in **S31–S33**, the G1 family in **S12**; the reading below is stated more weakly in **Supplementary S11, section 13.3**).

**We then ran the clean test** of G3's epoch-budget series on **one machine with early stopping disabled**, all four budgets present: C1 is **met**, C2 falls **outside** its band, and C3 **passes** on the mean ê/E it specifies but not per arm — see §3.1 and **Table S34b**; **the pre-declared verdict stays with the with-early-stopping series**. The judged criteria carry one message, **at the level of sign, not of functional form**: the selection bias moves with the amount of selection noise and with the number of selection opportunities, but not with whether the two splits are the same file — **the form itself was withdrawn** (§3.1). Two consequences do not follow: §4's configuration-level fact stands, while the stronger reading — that aliasing *adds* a measurable amount on top — does not. **The premium this paper measures is the premium of selecting on the reported split; aliasing is the configuration that guarantees it, not an additional term.**

On the shared 5-epoch grid the `val` and `test` peaks agree in only **17.5 %** of runs, and conditioning on that single observable splits the 40 runs: where they

agree the **median Δgap is lower** (mean **+0.13 pp**, median **−0.20 pp**, n = 7; **post-hoc**), where they differ it is **systematically not** (mean **+0.91 pp**, n = 33; permutation p = **0.009**, **post-hoc**; the median reading is **p = 0.0011**). **This is a partial predictor, not an explanation of the sign**; its limits — including the four of seven agreement runs whose gap is non-positive — and the per-run table and test are in **Supplementary S11**.

## 4. Finding 1: our own corpora

### 4.1 How common the aliasing is

In **our own packaged configurations** — the YAMLs we train from, a different object from a benchmark's own release — **6 of 14** give the reported number no independent held-out split: **AI-TOD, DOTA v1, DOTA v1.5, Mendeley face-mask, SFCHD** (five binding `val:` to `test:`) and **SHWD** (which binds `train` to `val`; **S1** row 13). Of the remaining 8, **4** keep the keys genuinely distinct and **4** have no `test:` key at all — the three groups account for all 14, and this is **a count over the configurations we hold, not a prevalence, nor the benchmarks' own releases** (audited separately, §6.1). Counting by file, **19 YAML files** alias the two keys, in five locations; on the archived manifest subset the ratio is **8 of 28**. **These three bases answer three different questions** — configurations (6 of 14), the files inside them (19), the archived manifest's rows (8 of 28) — **not three estimates of one rate.**

**For the corpora whose `val` and `test` are the same directory this is not contamination**: their image-level intersection with training is **exactly zero**, so the aliasing changes not *what* was evaluated but *which checkpoint* was reported, turning "selection contamination" from an ethical question into a **measurement** question. **SHWD is the exception and is not folded into that statement**: with `train` and `val` on one directory, its training and evaluation images are the same files. **It still counts among the six, for the reason the list is about: its `val` is not an independent held-out split either.**

### 4.2 The failure is produced by a tool, not by carelessness

A PASCAL VOC [21]→YOLO conversion script wrote two keys from one `images_dir`, so `train == val` appears **because the generator could not produce distinct values** (§6.1 records the same for Mendeley face-mask [22]). §10's first recommendation: make "are `val` and `test` the same source?" a **pipeline-level hard check** — human YAML review will not catch it.

## 5. Finding 2: the selection premium — magnitude, direction, and mechanism

**Evidence.** `eval_validity/selection_premium_local_20260916.txt`, reading **569 archived runs** from the local archive (the local version is the 2026-09-16 repair).

**Definition.** premium = **best-epoch − final-epoch** mAP50-95 **within a single run**. When `val == test`, "best" was selected on the reported split, while `last` and `last5` are endpoints **no selection rule can reach**. This is the checkpoint-selection term of the run-to-run variation studied in [23] (early stopping, data order) and [24] (benchmark variance across seeds). Being a maximum over a noisy curve, it also carries the upward bias of any selected extreme [13]: what those works average over as uncertainty, we measure as a reproducible component. **The two arms.** Every cell is a paired comparison in which the **baseline** arm follows the published convention and the **strategy** arm is the intervention under test — in the cells reported here, raising the peak learning rate from **0.001 to 0.005** with everything else held fixed (batch 32, imgsz 640, SGD; YOLOv12n with the shape-aware IoU). The two arms run the same code and the same data.

> **Table S3** → Supplementary §S8.

### 5.1 Direction: reported in two tiers, and the second tier's sign varies by corpus

The direction claim **must be reported in two tiers**: "the convention understates the effect" is wrong, and external checking refuted it (§5.4, II). **Endpoint convention (frozen):** `best - last`
**within a single run**, both endpoints reported for every cell; §7.2–§7.3 quantify the other convention.

**Tier 1: the selection premium is positive in all four cells and the baseline arm gains more**, consistent with the selection-bias literature, not new in kind; **on the monitored split that positivity is near-definitional** (`best` is that curve's own argmax), and **the finding we report is the paired arm difference and its test side (§8.6)**. Pairing brings **no** variance benefit: the paired t of 5.74 and the unpaired Welch t [25] of **5.68** agree — at $\rho = 0$, **standard arithmetic rather than an empirical finding of ours** [24]. **Permutation checks** agree with the t-tests at each cell's attainable floor (**Supplementary S12**).

**Meta-analyses of published results have long quantified this optimism**: development-set early stopping is measured to *overestimate realistic model performance*, by as much as **18.0 %** accuracy [26]; and a large Kaggle meta-analysis finds *little evidence of substantial overfitting* [27] — complementary rather than contrary, since its mechanism is holdout **reuse** and ours needs none.

**Metric sensitivity.** On `metrics/mAP50(B)` the mean direction holds in all four cells, but `smoke2sf` loses both its significance (**+0.063 pp**, p = 0.54) and its sign unanimity (**6+/5−**, not 11+/0−): **the mean direction is metric-robust, one cell's significance and sign unanimity are not** (full mAP50 columns and both mAP75 columns — the mAP75 `val` side is a **5-epoch grid**, so a rate computed from it is **upward-biased** — **Supplementary S12**).

**Table 1.** Tier 1 — each arm's **peak−final gap** and the **paired difference** between arms, by cell; the gap is a within-run maximum (§5.1), so the finding is the paired difference. `n` counts paired units (**eleven**: ten `--shuffle-seed` runs plus one unseeded — Supplementary S12); both differences hold at n = 10. **A different run set from the transfer measurement in §8.6**: these are the 569-run local archive under the published (aliased) protocol, while that measurement's premiums come from the clean three-way runs — the two tables' premiums are different quantities, not two readings of one. Source: `verify_two_critiques_20260916.txt`; the `dota15` row's n = 10 escalation is in Table S28.

| Cell | n (paired units) | peak−final gap, base | peak−final gap, strat | **paired diff** | sd | t | p | perm. *p* (floor $2/2^n$) | 95 % CI |
|---|---|---|---|---|---|---|---|---|---|
| smoke2sf (strong) | 11 | +0.799 | +0.382 | **+0.416** | 0.240 | 5.74 | $1.9\times10^{-4}$ | 0.00098 | [+0.255, +0.578] |
| shwd2sf (corroborating) | 11 | +1.038 | +0.512 | **+0.526** | 0.207 | 8.41 | $7.6\times10^{-6}$ | 0.00098 | [+0.386, +0.665] |
| a2d15 (directional probe) | 10 | +0.534 | +0.262 | **+0.272** | 0.183 | 4.69 | $1.1\times10^{-3}$ | 0.00391 | [+0.141, +0.403] |
| dota15 | 10 | +0.94 | +0.49 | **+0.4 | 0.1 | 7. | 4.9× | 0.00 | [+0. |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| (within-domain) | 1 | 2 | 49** | 96 | 23 | $10^{-5}$ | 195 | 309, +0.589] |

**Table 2.** Tier 2 — the paired difference Δgap by corpus. Δgap > 0 means `val` understates the strategy arm's effect. **Δgap is the split-gap arm difference**, not the selection effect: it decomposes into a difficulty gap and a selection term, and **Δgap's sign is not the selection term's sign** (**Fig. S1**). Equal split difficulty across arms is an **assumption we state**; the selection contribution is the **selection-action contrast** `Δprem_val − Δprem_test`. **`Reading` names Δgap's sign, not the selection term's** (`smoke2sf`: Δgap **+0.461**, selection **−0.580**). Sources: `xeval_analysis_20260916.txt` (the two `val`-understating cells) and `x1_tier*` / `teval.csv` (the two `val`-overstating cells).

| Corpus | n | Δgap (pp) | p | 95 % CI | Reading (**Δgap's sign**) |
|---|---|---|---|---|---|
| shwd2sf | 10 | **+0.779** | 0.0053 | [+0.297, +1.261] | `val` **understates** |
| smoke2sf | 10 | **+0.461** | 0.0263 | [+0.068, +0.855] | `val` **understates** |
| p_aitovis | 10 | **−1.226** | $1.3\times10^{-8}$ | [−1.370, −1.081] | `val` **overstates** |
| p_vistod15 | 10 | **−3.315** | $6.5\times10^{-9}$ | [−3.676, −2.953] | `val` **overstates** |

> **The split of Δgap is recorded for two of the four cells**: difficulty **+0.759** / selection **+0.020** (`shwd2sf`), and difficulty **+1.041** / selection **−0.580** (`smoke2sf`) — in the second the selection term is **opposite in sign to Δgap**, so the sign does not carry it, and **both intervals include zero** (**Supplementary S9**, Fig. S1). The two `p_` cells' `prem_test` was **not recorded at that time**: their per-epoch `test` curves were evaluated on 2026-09-30, so nothing is imputed and the values are now recorded (**Supplementary S12**, Table S38).

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> **The four rows below are an escalated base, not a sample**: the two `p_` cells reached n = 10 because their n = 3 Δgap had been flagged, so the 2/2 balance is a property of the escalation rule; the other nine cells are **descriptive** (**Supplementary S12**). This cell is also the one where **3 runs** changed training path silently under GPU memory pressure, and why we do not call it clean is in **§10, defect 10**.

*(Third §5 table, S5 → Supplementary §S8.)*

**Tier 2: on the same three-way corpora the paired difference Δgap varies in sign.** The raw `M_val(ê) - M_test(ê)` is **not identifiable**; the paired difference removes the difficulty term **under the equal-difficulty assumption** (Table 2) —

**Δgap > 0 ⇒ `val` understates**. **Its effective base is four cells, not thirteen**, and an **escalated base** rather than a random sample, for the reason **Table 2's note** gives; **"2 understate / 2 overstate" is therefore a statement about these four cells, not a rate for the family**. **No prospective power analysis was registered**; the scan is in **Supplementary S12**.

**Both escalations met the criteria fixed in advance.** `dota15` **+0.449 pp** (**10+/0−**); `p_aitovis` **−1.226 pp** and `p_vistod15` **−3.315 pp** (**0+/10−**), permutation p = **0.001953**, the attainable floor at n = 10. The `best.pt` endpoint was identified, not assumed: on `test` it reproduces the archived `sio_b_results.csv` to **0.0046 pp**.

**The formal claim:** the convention systematically shifts **arm-to-arm comparisons** (Tier 1: four cells, CIs excluding zero); **the direction of that shift varies by corpus** (Tier 2: 2 understate / 2 overstate on the four-cell base) — **neither "a general understatement" nor "no systematic direction"**. BH at q = 0.05 **now leaves three cells significant** — `p_vistod15`, `p_aitovis`, `shwd2sf` [28], **reversing what we reported before the escalation**. **The two families are not the same test**: the **13-cell scan is descriptive**, the **confirmatory family is the four escalated cells** (declared before their runs, §3.2) — the direction statement rests on those four.

**Scope (cross-framework boundary).** The convention this paper audits belongs to the **YOLO lineage — Ultralytics YOLO [16] and YOLOX [29]**: at pinned commits, Detectron2 (`TEST.EVAL_PERIOD = 0`) [30], MMDetection [31], DETR [32] and PaddleDetection [33] **report the final epoch instead** (§6.5) — that pipeline family's convention, **not the field's**.

### 5.2 Two versions must be disclosed side by side

**Two archives, two versions of this table.** The main text uses the **local archive** (569 runs, n = 11); the pod root (347 runs, n = 10) is a **Supplementary S12** cross-check agreeing in sign and magnitude, not in value.

### 5.3 The dota15 cell: now reproducible, and one wording defect exposed

**This cell was not reproducible from the release as first written, and the repair exposed a wording defect** (both disclosed, §10 row 2). The repair, the value-by-value agreement and the defect are recorded in **Supplementary S12**.

### 5.4 Two external critiques, checked (one refuted, one judged non-identifiable)

**Supplementary S12** carries the two external critiques in full. **We used to write** that the gap was simply unidentifiable; the external check **corrects this to** the paired-difference form, which cancels the difficulty term under the pairing assumption stated in **§S11**. The argument and the recomputation are in **Supplementary S12**.

### 5.5 The same component in third-party public logs

**The same quantity appears in other people's logs — a convenience sample with no sampling frame, not a field-level rate**: 20 logs from 6 sources found by repository **search**, median **+1.04 pp**, max **+5.65 pp**. **The sign is not the informative part**: the premium is a within-run maximum minus that run's own final epoch, so it is **non-negative by construction**, and the sample's content is its **magnitude**. The queries, the rest of the frame and the reason public logs **cannot establish aliasing** are in **Supplementary S12**.

**Five limitations**: ① the two cells escalated in §5.1 reach n = 10 with **local** runs and test-side evaluations, and the remaining 9 cells' `test` side was closed on 2026-09-30 on one machine; ② `prem_test` is **best − last**, while the per-epoch `test` curve now exists for **all thirteen** corpora (**Supplementary S12**, Table S38); ③ `last.pt`'s val recomputation differs from the final `results.csv` row by ~0.02 pp; ④ this section does **not** touch §6's 19-benchmark census; ⑤ the aggregate is a **ratio of small means** — `prem_val` spans **+0.605 to +1.469 pp** across the five arm-cells (Table S20) — so a change of a few hundredths of a point in the numerator moves the rate by tens of points.

### 5.6 Why Δgap's sign varies by corpus: three of five candidate mechanisms are refuted by the data

**Three of five candidate mechanisms are refuted, and the mechanism itself is not identifiable** — the sign of Δgap is a sum of same-order terms, so no single mechanism can be read off it (§S11, Remark 3) — **but the paired**

**quantity is, and is what we report.** The examination, the per-cell components and the exact identity are in **Supplementary S12** and Table S8–S10.

## 6. Finding 3: how general this is across public benchmarks

**Evidence**: `benchmark_split_audit.md`, a line-by-line audit of 19 benchmark rows, plus `split_units_19rows_20260916.md` (recomputation script; no hard-coded percentages).

**The nineteen audited benchmarks**, each read from its own release or dataset paper: COCO [34], PASCAL VOC [21], Objects365 [35], Open Images v7 [36], DOTA v1.0 and v2.0 [37], VisDrone-DET [38], AI-TOD [39], UAVDT [40], xView [41], DIOR [42], NWPU VHR-10 [43], SHWD [44], SFCHD [45], MAFA [46], Mendeley face-mask [22], WIDER FACE [47], CrowdHuman [48] and D-Fire [49] (**per-unit markings**: Supplementary S1) — **the configurations this archive could read, not a sample of the field.** The question is **evaluation validity**: dataset-level bias auditing [50] and classifier-accuracy estimation [51] ask whether a reported number measures what it is taken to measure, but neither audits a release's configuration binding.

### 6.1 The result under four units (single-rater judgements)

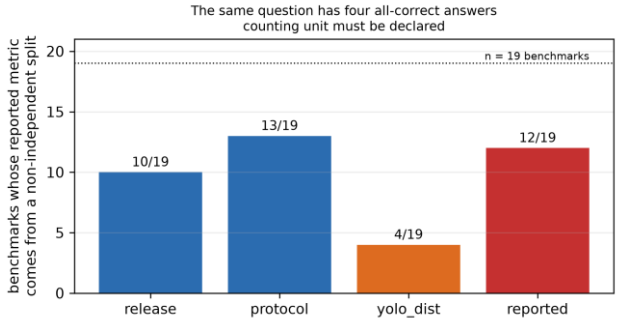



**Fig. 1.** The four counting units on the same nineteen benchmarks — the **printed single-rater** counts (10/19 `release`, 13/19 `protocol`, 4/19 `yolo_dist`, 12/19 `reported`); the re-verified tallies are Table 3's second column.

**Table 3.** The four counting units, with denominators named (§3). **Co-headline (2026-10-10):** the same four units were re-coded blind by **five current-generation coders** (κ **0.824**; **Supplementary S12**), and re-verifying every row first-hand changed **30 cells** — so the table prints **both tallies**: the frozen single-rater counts, and beside them the same four units **re-tallied from the marking table after the 2026-10-09 first-party correction record**, under the **same per-column rule** (the value-to-count mapping is in **Supplementary S12**). Source:

`split_units_19rows_20260916.md`. **Rows marked `n_a` enter the denominator, not the numerator**; **alternative readings of the same rows** are **12/19** for `release`, **12/15** the 19 less the four **undecidable** (`n_a` is two rows), and **13/19** for `reported` under the COCO-`val` reading. **The `clean` row is a reading**: the audit counted **5/19**, a later pass moved **two**, so this table reads **3/19**. **The `reported` row's two denominators**: `12–14 of the 19`, and `12/15 = 80 %` over the fifteen neither `n_a` nor undecidable (**4 rows are evidence-insufficient**). **Both ranges span two readings, not intervals**, on a **single rater's** judgement. **Independent re-coding reproduces it only at 31.4 % (Round 7, coder agreement κ 0.836) and 42.2 % (Round 8, κ 0.811) of the printed marks** (Supplementary S12). A **model-assisted, non-blind second coding** is **reliability evidence, not a replacement** (**Supplementary S12**).

| Unit | non-independent: printed (single rater) | non-independent: after first-party re-verification | of decidable rows: printed → re-verified | undecidable: printed → re-verified |
|---|---|---|---|---|
| **release** (the benchmark's own release) | **10/19 ≈ 53 %** | **14/19** | 10/19 → **14/19** | — |
| **protocol** (the official protocol) | **13/19 ≈ 68 %** | **14/19** | 13/19 → **14/19** | — |
| **yolo_dist** (the generic YOLO distribution package) | **4/19 ≈ 21 %** | **2/19** | 4/15 ≈ 27 % → **2/17 ≈ 12 %** | 4 → **2 rows** |
| **reported** (the layer actually reported) | **12–14 of the 19 ≈ 63–74 %** | **10–11 of the 19** | 12/15 = 80 % → **10/16–11/17 ≈ 63–65 %** | 4 → **2–3 rows** |

> **What a count here is, and what a "yes" means.** Every row is a **single rater's judgement**
> (§6.1), so the counts are printed as counts — the percentages beside them are a reading aid, not a
> rate estimated from independent coding. A "yes" under any
> unit means the reported number is **not a sample from an independent split**; it does **not** mean

> the benchmark is broken or the experiment dishonest, and it does not license a generalization claim.

**These 19 rows were coded by one rater** — 28 literal judgements (**§10**, defect 5). The five blind coders **independently reach 19 of 30 (63.3 %)** corrected cells, eleven unanimously, while **26 of 76** cells stay **undecidable**; a **non-blind, model-assisted** pass agrees at κ **0.874** and **moves two rows** (**Supplementary S12**). `reported` means **the number the literature prints**: it is printed as a range, **12–14 of the 19 ≈ 63–74 %**, with the **80 %** column the lower end over the **12/15** decidable rows, and `n_a` rows counted in the **19** but not the numerator (**Supplementary S12**).

### 6.2 "Same directory" has four levels, and the audit reported only the narrowest

The levels, the rows each admits and the resulting shift are in **Supplementary S12**, Table S12.

### 6.3 We do not reproduce the original audit's two numbers, and we say why

one rests on the narrowest reading of "same directory", the other on a stale snapshot; both recomputations are in **Supplementary S12**, Table S13.

### 6.4 One judgment the evidence does not support (corrected)

**One classification correction is recorded in Supplementary S12** (D-Fire moves to the clean group).

### 6.5 Across frameworks: is the convention specific to YOLO?

**Among the pinned default configurations we checked, only the YOLO lineage does this by default.** At pinned commits Detectron2 (`TEST.EVAL_PERIOD = 0`), MMDetection, DETR and PaddleDetection **report the final epoch instead** (file-by-file evidence, 25 configuration files: **S12**, Table S14; **a targeted sample at pinned commits, not a census**, so "the convention" means that family's, not the field's). A **cross-lineage realization test** — a Faster R-CNN (ResNet-50 FPN) detector, a family that **reports the final epoch by default** — on VisDrone under the same three-way split, at **six seeds**, is **positive in all six runs, with a 95 % interval excluding zero** (**68.2 %** realized; **+0.0809 pp** mean `prem_val`, 95 % CI **[26.9 %, 123.5 %]**) (values, interval and per-run table: **Supplementary S34**): the mechanism is **not confined to the YOLO lineage in this instance**. A **second detector family** (RetinaNet, both corpora) keeps the **direction** but not the strength

(**Supplementary S34e**). A second corpus, same protocol, same direction: **6/6** and **12/12** runs on `val`, **5/6** and **11/12** on `test`. This is **one detector, one protocol, two corpora, one setting — not a generalisation beyond the YOLO lineage** (**Supplementary S34c**). Its **68.2 %** is a rate over **six runs** on this probe's own split and is **not comparable** with §8.6's realization rate: different protocol, different denominator.

## 7. Finding 4: what "significant" means, and how the endpoint convention changes it

### 7.1 With n = 3 a permutation test cannot reject in principle

**A power note for the cells that remain at n = 3.** Recovering each Tier-2 cell's sd from its own printed 95 % CI gives **0.20–0.67 pp**; at n = 3 the required effect is **0.65–2.19 pp** at α = 0.05, two-sided — **at or above every Tier-1 effect here** (+0.27 to +0.53 pp) — and 0.2–0.7 pp at n = 10. The nine cells at n = 3 are therefore **not testable at this design**, and are reported descriptively only. **A prefix-n table (n = 3…10) and the derivation with the non-central-`t` factor (`3.26·sd`) are in S12**; the permutation floor **$2/2^n$** binds at n = 3, not the *t* test.

**For a sign-flip permutation test on n pairs the smallest attainable two-sided p is $2/2^n$**: at n = 3 that is **0.25**, so such a cell has **no resolution at this n**; **after §5.1's escalations no headline cell (Tables 1–2) is n = 3** (**the derivation and each n's floor: Supplementary S12**).

### 7.2 The two clean-protocol cells clear significance by n = 6–8

**The weaker endpoint reaches *t*-significance at n = 6 on `shwd2sf` (p = 0.0164) and n = 7 on `smoke2sf` (p = 0.0400)**; the **exact permutation — floored at $2/2^n$ (0.25 at n = 3) — follows** one step behind on `shwd2sf` and none on `smoke2sf` **(n = 8 on both `selected` readings; Table S42 gives every prefix n, both endpoints). The clean-protocol seed extension that reaches n = 10, and the per-cell numbers, are in** Supplementary S12** and Table S15.

> **First-report attribution (a convention we hold to).** Three things here belong to the companion

> paper and are cited, not claimed: the two Δ values (**+1.434 / +0.507 pp**), the three-way remediation

> protocol itself, and the pre-registered multi-target replication (§9). What this paper reports first is

> **the selection premium (§5), the 2 × 2 realization of that premium on a disjoint split (§8.6–8.7), and

> the formalisation of the realization rate (§3.1)**. The full statement — including how the companion is

> named to the editor while it carries no identifier — is in **Supplementary S12**.

### 7.3 A cross-archive endpoint conflict on the same run (investigated — and itself a disclosure)

**One run is recorded with two different endpoints in two archives** — which checkpoint is recorded was never specified: **Supplementary S12**, Table S16. Taking the final-epoch entry (67.12) gives Δ = **−1.490**, p = **0.0141**. The best-epoch entry (70.06) reads Δ = **−0.902**, p = **0.2931**, **but had been computed with the run's `last` reading**; corrected **2026-10-02** it becomes Δ = **−1.490**, p = **0.0141**, as released. So under the **`best`** convention of §5 this cell is **significant**, and nothing printed here moves.

## 8. Finding 5: how far the selection premium transfers to an independent split

### 8.1 The protocol is the companion's; the measurement under it is this paper's

The three-way remediation protocol is **defined and reported in the companion paper** — which first reports that **the effect reproduces in direction while its magnitude falls** — so this section **cites it** rather than presenting the replication. **In this paper's own terms it is three things**: a `val` carved from the pool so that training never sees it; the target corpus's own held-out `test`, with a filename-level check that its images do not intersect training; and an endpoint **defined in writing before any run**. Its **FROZEN-HASH** stamp is stated in **§9**; the registration **text** itself is the companion's and is **author-side**, so this paper carries the stamp and the readings (**Tables S24–S27**), not the document. What this paper measures

**under** that protocol is the quantity §3.1 formalises: **how much of the `val`-selected premium survives on a split no selection rule can touch**, and that measurement is this paper's.

### 8.2 The protocol's results on the companion's corpora: transferred

The protocol's results on the companion's corpora are the companion's first report. What this paper adds under the protocol is the measurement of §8.6–§8.7; the transfer (2026-09-21) is recorded in the module-transfer note of that date.

### 8.3 A direct measurement of data-order replication (not seed variance, and not the selection claim itself)

**A direct measurement of data-order replication — reproduced inside this setting**: adding replicates that permute the training file order lowers the mean by **9 %–29 %**, each figure taken **against that cell's own n = 6 mean** and not against the pooled headline effect, the **seed-budget discipline** measured rather than assumed ([52]'s setting, §8.4). **What these replicates are.** The per-run randomness is a **deterministic permutation of the training file order** (`--shuffle-seed`, the number in the run's name), **not** the framework's `seed` argument (**Supplementary S12**, section 9.3, with the two corpora whose replays are identical across `--seed` values). The quantity is therefore **data-order replication**, not seed variance: the 9 %–29 % bounds the **file-order** component of the reported mean, **not** the variance between independent training seeds. That best-of-K selection inflates a reported score is **not new** (§8.6 positions the closest concurrent work, a dose–response over 2,047 datasets), so **what is measured here is the magnitude under a protocol that selects cells for significance first and expands seeds afterwards** (design and recomputation check: **Supplementary S12**, Table S18). **The title is deliberately the weaker, sample-scoped claim**: with three initializations the interval is wide, and extending to five does not narrow it (the gain SD reaches the data-order component's order), so the components are **consistent with, not established by**, these initializations.

### 8.4 The three variance components: the excluded one is the smaller in this sample — a sample-scoped ordering, not an established one

**The three variance components, and the excluded one is the smaller in this sample**: the single-run data-order component is **0.515 pp**, against an initialization component of the same order — the **same ranking** as the shuffle-order result of [52] (**Supplementary S12**), so we **reproduce the ordering inside this setting** rather than discover it. What was held fixed, and why this is **sample-scoped** — at five initializations the gain SD moves **0.146 → 0.439 pp**

and its interval **overlaps** the data-order magnitude, so the two are of the same order rather than one clearly smaller — is in **Supplementary S12** and Table S19.

### 8.5 Why "same-pipeline pairing" is a necessity, not a habit

**Pairing is not neutrality**: the paired design covers the **budget and data** axes only, and **not** the optimiser's (why, and the counterexample: **Supplementary S12**).

### 8.6 The premium's realization rate: the claim-level interval includes zero ([−17.3 %, 53.2 %], P = 0.118); the point estimates below are descriptive (arm-cell median 17.6 %, five arm-cells, n = 10; run-weighted 23.3 %)

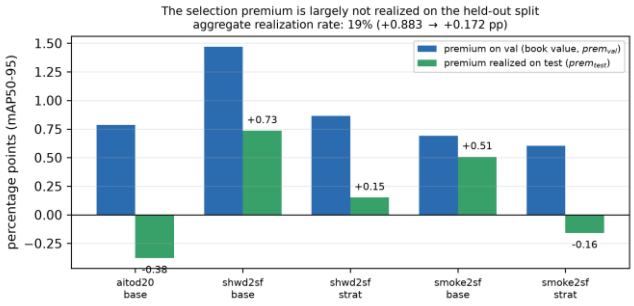



**Fig. 2.** The `val`-selected premium against what it realizes on the disjoint `test` split, per arm-cell; the title's aggregate is the plotted pool's — **§8.6** names the pool Table 4 uses.

**What this section claims, and what it does not.** The rate below is the realization of the `val`-selected premium on a disjoint split **for the five arm-cells measured here, under the companion's protocol** — **not a constant**. The escalated cells reach a **pooled 55 %**, but the two batches are **not pooled** (different runs, different protocol, §8.1). "The convention realizes about a quarter of the `val` premium" would be false; "in these five cells, about a quarter survives" is what we measured.

**How much of the gain selected on `val` survives on `test`?**

**Table 4.** The premium selected on `val` versus what that same selection action realizes on the disjoint `test` split, per arm-cell (**an unbalanced pool** — two corpora contribute both arms, `aitod20` only its baseline — so the arm-cell is the unit). Source: the **per-epoch `val` curves** (`results.csv`, for `prem_val`), `xeval_20260916/matrix.csv`, and this revision's two batches (`x4_teval.csv` on A, `x4fill_20260925.csv` on B, for `prem_test`). **The**

**§10 memory-pressure bound (defect 10) does not arise here**: its hits are in §3.1's clean batch and §5.1's `p_vistod15`.

| Cell | Arm | n | `prem_val` | **`prem_test`** | **realization rate** |
|---|---|---|---|---|---|
| shwd2sf | baseline | 10 | +1.469 | **+0.736** | **50 %** |
| shwd2sf | strategy | 10 | +0.865 | **+0.152** | **18 %** |
| smoke2sf | baseline | 10 | +0.691 | **+0.509** | **74 %** |
| smoke2sf | strategy | 10 | +0.605 | **−0.157** | **−26 %** |
| aitod20 | baseline | **10** | **+0.766** | **−0.218** | **−28 %** |
| aitod20 | **strategy** | **10** | **+1.111** | **−0.543** | **−49 %** |

**†**

> **The inclusion rule, stated as a rule.** An arm-cell enters the pool **iff its `test` side reached n ≥ 5** (`work/item2_final_v2_20260925.py`), applied to every cell alike; **exactly five met it at the 2026-09-25 freeze** (50 runs). The `aitod20` strategy row **qualifies under the same rule but completed after the freeze**, so it is printed here as the sixth row and kept out of the primary pool — **the `†` row, and the only one**; **both readings are reported and both intervals cover zero** (Table S20's note, §8.6b).

where `prem_val = V_max − V_final` (**selecting on val**) and `prem_test = T_best − T_final` (**realized on test**).

**The three bases**: run-level `[8.4 %, 37.4 %]`, `P = 0.002`; five-cell `[−17.3 %, 53.2 %]`, `P = 0.118` (claim-level, **zero inside**); eleven-cluster `[19.3 %, 60.4 %]`, `P < 0.001` (a bootstrap tail proportion **at the 20,000-resample resolution**, i.e. below the floor this procedure can print).

**Aggregate** (runs weighted equally, n = 50): `prem_val` **+0.879 pp** → `prem_test` **+0.205 pp**. **At the claim level this is not significant**: the **claim-level interval** is the cell-level, five-cluster one, **[−17.3 %, 53.2 %]**, **P(rate ≤ 0) = 0.118**, zero inside. Descriptively, arm-cell median **17.6 %** and run-weighted **23.3 %** (span **2/5** negative: Table 4). On a difference scale the eleven clusters give **−1.367 pp**, 95 % CI **[−1.59, −1.15]** (five-cell denominators **+0.605 to +1.469 pp**). Re-aggregated: run-weighted **36.9 %**, arm-cell-equal **39.3 %**, 95 % CI **[19.3 %, 60.4 %]** — **completing the cell, not the weighting, is what moves it**: at n = 10 the three weightings agree, so **the fragile axis is pool composition**.

**Leave-one-out 9.77 % to 34.16 %.** In two arm-cells the selection action **costs** rather than fails to pay off (`smoke2sf` strategy, `aitod20` baseline); the

escalated cells of §5.1 are **not pooled with these five** (different runs, different protocol). The closest concurrent work [12], the winner's-curse check, the withdrawn σ√(2 ln E) form and each cell's corpus pair: **Supplementary S12**.

(the `best.pt`-versus-argmax residual, our withdrawn attribution and why the remainder is **unidentified**: **Supplementary S12**.)

The monotonicity is a correlation, not a causal claim, and at the arm-cell unit the nine-cluster reading is ρ = **−0.433** (p = 0.244) — see **Fig. 3** and **Supplementary S12**: the n = 80 are the 81 runs with a measured `test` side less the one whose `prem_val` is 0. It has a named precedent: [53] gives validation overfitting (VO) its name and bounds the expected best observed loss by E[min L] ≥ v* − s_m√(2 log(Nq)), an optimism growing with the number of selection queries Nq.

**Recomputation status.** The readings were originally **not locally recomputable** (the archive lacks 6 of 7 corpora; **§10, defect 5**); a GPU **2 × 2 evaluation** on **41** three-way runs of **YOLOv12n [54]** × {`best`, `last`} × {`val`, `test`} makes them so, **retraining not**, reproducing the archived `best × test` values to within **0.0047 pp** (median **0.0021 pp**) — the endpoint rule's positive control (**Supplementary S12**).

**Three limitations**: ① in the **41-run epoch sweep** only the two headline cells reach n = 10; `aitod20` contributes one run of 30 epochs and 6 checkpoints where others contribute **20** (hence 806, not 820), its Table 4 arm-cell a separate n = 10 batch; ② **closed: every retained checkpoint is now evaluated**, the *stability* reading is narrower than the grid; ③ κ is relative to each run's own final epoch, not causal. The 806-row evaluation is archived (`xeval_perepoch_20260918/`, with a SHA-256 list).

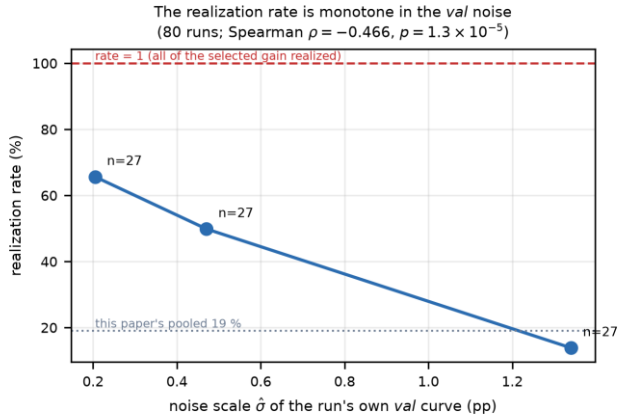


**Fig. 3.** Realization rate against the noise scale $\hat{\sigma}$ of the run's own `val` curve, over **80 runs** in three terciles of **27 / 27 / 26** (nine-cluster Spearman $\rho = -0.433$, $p = 0.244$; the run-level reading is −0.466); dashed: rate = 1, dotted: §8.6's pooled reading.

**Pool composition — and why the claim does not turn on it.** Table 4 prints **six** measured arm-cells; adding the sixth (**`aitod20` strategy**) moves the run-weighted rate **23.3 % → 8.7 %** and the median **+17.6 % → −4.2 %**. **The inclusion rule is a rule, not a per-cell choice**: an arm-cell enters **iff its `test` side reached n ≥ 5**, and **exactly five met it at the freeze**, where all three weightings coincide (**23.26 %**). The sixth row **qualifies but completed later**, so it is the sensitivity. **Every claim is interval-level and any scope gives the same verdict** — five-cell **[−17.3 %, 53.2 %]** (P = 0.118), six-cell **[−4.9 %, +21.3 %]**, eleven-cluster **[19.3 %, 60.4 %]** — the cell-level intervals **both cover zero**, so **no conclusion depends on the pool boundary**.

### 8.7 The per-epoch view: the realization rate is not a constant

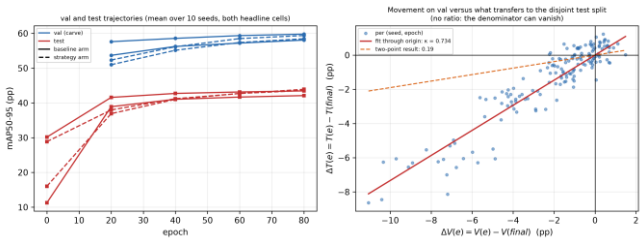



**Fig. 4. Left**: mean `val` and `test` trajectories over 10 seeds, both headline cells (baseline solid, strategy dashed). **Right**: per-(seed, epoch) `val` movement against what transfers to `test` — the ratio is not taken (its denominator can vanish); the fit through the origin is **$\kappa = 0.708$** ($R^2$ **0.937**, run-clustered 95 % CI **[0.671, 0.745]**; the title's **0.734** is the earlier five-epoch grid).

**One metric self-correction first.** Our first version used `r(e) = ΔT(e)/ΔV(e)`, producing values such as **−860 % and +1031 %**: at late epochs `V(e) ≈ V(final)`, the **denominator vanishing** — **an ill-conditioned metric, not a finding.** We report only the non-dividing `ΔV(e) = V(e) − V(final)` and `ΔT(e) = T(e) − T(final)`, summarized by the through-origin slope κ of ΔT on ΔV.

> **Tables S21–S23** → Supplementary §S8.

**Slope κ** (how much of the movement on `val` transfers to `test`). On the **complete curve** —

**n = 765 (seed, epoch) points over the 41 runs** — **κ = 0.708, $R^2$ 0.937**, with a run-clustered (sandwich) 95 % CI **[0.671, 0.745]**; the within-run estimator gives **0.698**. **Two controls (Table S43)**: a per-run κ distribution, and a free-intercept fit whose **intercept is indistinct from zero** — so the through-origin constraint is **supported, not assumed**. Across the curve ΔV reaches

**−8.1 pp** and about **71 %** transfers: **early-training progress is real progress.** §8.6's realized share measures the **selection-relevant sliver** near the peak, where ΔV is of order **±0.2 pp** — so

**"most of the movement on `val` transfers" and "the little extra gained by selecting near the peak does not" are both true**, and the premium is the residual of the transfer rate. **And κ is not constant across arms either — in the direction the selection account predicts**: **baseline κ = 0.600** ([0.581, 0.619]) against **strategy κ = 0.767** ([0.735, 0.800]), difference

**+0.168, 95 % CI [+0.130, +0.203]**.

**A second result, and a self-correction the curve forces**: the sign-of-Δgap claim we wrote from four sampled epochs is **too strong for one of the two cells** (on the complete curve `smoke2sf` starts negative) and is **narrowed here**; per-checkpoint values: **Supplementary S11**.

**The limitations of this section are in Supplementary S12.**

## 9. The registered multi-target replication: reported in the companion paper

**First report, criterion and verdict stay with the companion paper; nothing here is a claim of this paper.** To read its registered table: ① it is a **three-way split** (train / selection / report); ② it ran under the protocol then in force, where the **`val` side aliases the `test` half** — the one fact §7 takes; ③ its criterion was frozen **before any run existed** (FROZEN-HASH `6a7eee7b3e34b15ce5adcba14cf7ea36`) and its verdict entered is **"criterion not met"**, deliberately **not** "hypothesis refuted" — the companion reads the outcome as **"direction replicates, size does not"**. Record: **Supplementary S24–S27** (source of truth: the companion's archive).

## 10. Our own defects (disclosed, not hidden)

| # | Defect | Handling |
| --- | --- | --- |
| 1 | The clean-protocol results | **We do not point at a |

| | | |
|---|---|---|
| | document **cites a ledger file that does not exist in the sense claimed**: neither `_r10_final_stats.txt` nor the "real log" `_r10_stats_0403.txt` holds **any** headline value (6 probes each, **0/6 hits**) ⇒ **no surviving stdout ledger** | replacement file** (renaming would manufacture a new false citation); the text says **"no stdout ledger"**, and the p-values are recomputable from `pvalue_table_local.py`. Detail: §S12 |
| 2 | ~~the dota15 cell of `selection_premium.py` was not reproducible~~ → **fixed** (stale root directory; the matcher assumed a seed suffix the run names do not carry); it now **reproduces M3's +1.12/+0.95/+0.93 value by value** | Stated in §5.3; the "3 seeds" note was **retracted on 2026-09-21** — the third run's seed is s42, the campaign's fixed shuffle seed. Detail: §S12 |
| 3 | **`sio_b_results.csv` has no written endpoint convention**: `t2_mask2mende_base100_s42n` is recorded as final (67.12) in one archive and best (70.06) in another | Stated in §7.3, with the consequence of each reading (significant versus not) |
| 4 | The two headline scripts **originally depended on a rented pod** (`import paramiko` plus a plaintext password) — a paper about evaluation validity whose own analysis is not reproducible from the release | **Fixed**: both now **read the local archive**; the main text uses the local versions and lists the pod versions as cross-checks |
| 5 | **One script now recomputes the σ decomposition** — the data-order and initialization components from the archived per-run values, each asserted against the printed number — but it can only **cite** the augmentation component, whose record holds an SD and no per-run values; **no frozen effect-size rule file**; **no script generates `benchmark_split_audit.md`** (hand-written, 28 judgments as literals, though its counts are recomputed) | The first is closed by script, with its one cited (not recomputed) component named; the other two stay labelled "manual / not archived" |
| 6 | The local archive **has no copy of 6 of the 7 corpora**, and of the three-way carve it holds **only the manifest** | **Partially repaired (2026-09-16, GPU run)**: the 41 runs' per-epoch curves, |

| | | |
|---|---|---|
| | (`split_defs_20260916/`), not the other carve directories | `sio_b_results.csv` and the **2 × 2 evaluation matrix** (164 evaluations) are now local; **retraining remains non-recomputable** |
| 7 | The audit document has **internal inconsistencies** (nine items, found by our own census) | Each adjudicated: **3 misreadings by the inventory, 4 real defects, 1 convention risk, 1 classification correction (§6.4)** |
| 8 | Our own corpora contain `val == test` | **That is this paper's argument**, not an embarrassment to be hidden |
| 9 | **Our first per-epoch report claimed curve-wide stability from a four-point sample**; the complete curve overturned it for one of the two cells (§8.7) | **Corrected in place, with the retracted sentence quoted**: the claim is **narrowed**, not deleted, and the grid's apparent uniformity is named as partly an artefact of where it sampled. Detail: §S12 |
| 10 | **Memory pressure silently changes the training path**: under GPU memory pressure Ultralytics switches implementation **without changing the recorded configuration** — a batch reduction, or a fallback of label assignment to CPU that can fork the trajectory — and case (a) is **invisible in the standard artifacts**. | **Disclosed, not retracted; we do not call the affected batches "clean".** Configuration-level hits: **2** places, **4** runs. **A reproducibility residual, not an event size**: same-config repeats are **bit-identical** (**0.000 pp**, four pairs, same machine and day), while the quantity a bound would be about — archived run against clean re-run — is **0.000 / 0.020 / 0.150 / 0.000 pp** (**Table S41**), so **≤0.05 pp** holds only inside the clean E = 50 batch and is **not extrapolated**; without a no-event control the |

event effect is **unidentified** (such a control is itself **not reproducible across dates**, **0.440 pp**), and elsewhere it sits inside the zero-event-control spread. The one clear case is ΔVmax **+0.769 pp**. Event counts, per-run consequences and the log fingerprints are in **Supplementary S12 (defect 10)**.

## 11. Recommendations

**Four checks before reporting, each with the artifact that discharges it.** ① Are `val` and `test` the same source? Make it a **hard pipeline check** on every dataset YAML (§4.2); the output is the resolved paths. ② If you must select on the reported split, say so and report a selection-free endpoint — **the `final`-epoch endpoint, the only selection-free one this archive measures** — alongside — the step **enlarges** the arm difference (§5.1). ③ State the test convention: below n = 6 a permutation test has no resolution; print the $2/2^n$ floor beside every p. ④ Declare the counting unit behind every count, as Table 3's header does.

⑤ **A plateau is a post-hoc candidate, not a rule we validate.** Weight averaging is established [55]; what is *measured here* takes the **mean of the ranked top-5 `val` epochs' readings** (a reading average, not a weight average), capturing **70 %** of the realizable gain against the argmax's **30 %**, at the cost of **0.73 pp** below the peak — a post-hoc reading of our own 40 runs, untested as an intervention. The per-rule table and the candidate mechanism are in **Supplementary S11**, Measurement 3.

## 12. Limitations

- The audit reads **split definitions** (YAML, README, paper text), **no datasets downloaded**.

- For several benchmarks the "generic YOLO copy" is **per-project**, not canonical.

- Under the `reported` unit, **4 rows (AI-TOD, DIOR, Mendeley, D-Fire) are evidence-insufficient**; we do not guess. Counts are **row-equal over the 19**, and the grading enters the **error bar, not the count** (**Table 3's second column**).
- Two evidentiary differences (pod versus local) are listed side by side.
- The census is a **literature/release survey**, weaker than our **self-measurement** (§3).
- The **difficulty/selection decomposition** (§5.6) covers **2 of the 13 cells** (the headline two); the **identity** behind it is checked on **all 13** (`work/gap_mechanism_20260916.txt`) — what is scoped is the decomposition, not the identity.
- The intervention axis the headline comparisons use is **`lr0` (`0.001 → 0.005`)**; two further orthogonal axes were run, one reproducing, as on `lr0`, the **corpus-dependent character** of the shift (**Supplementary S8**, **Table S40b**); the cross-lineage probe is
**one detector, two corpora, and incomparable magnitudes**.
- **Four disclosed uncertainties** bound our claims: the 19-row audit is **single-rater**; the log sample is a **convenience sample**; the GPU memory-pressure effect is **bounded, not estimated** (**≤0.05 pp** where same-config pairs can be compared, fresh replicates being bit-identical; cross-date comparisons carry a **0.440 pp** drift); and the §8.3 replicates vary the **training file order**, not independent seeds. **And, not as an uncertainty**: **no interval here folds into §8.6's rate**. **Supplementary S12**.

## 13. Conclusions

A count is meaningless until its **unit** is named: the same nineteen benchmarks answer differently under each declared unit (Fig. 1, Table 3) — **10/19** under `release`, **13/19** under `protocol`, **4/19** under `yolo_dist`, and **12–14 of the 19** under `reported` (whose COCO-`val` reading is **13/19**). **Re-tallied after the 2026-10-09 first-party correction record, under the same per-column rule and with these printed counts frozen, the same four units read 14/19, 14/19, 2/19 and 10–11 of the 19** (Table 3, Supplementary S12) — the two **layer** units rise, the two **package and report** units fall, and the **majority** conclusion is unchanged (the weakest recomputed **majority** is still **10/19**). Three findings stand. **The most robust number is the transfer slope κ (§8.7): early `val` movement does transfer to `test`, and the premium is its residual.** The

premium is positive in all four cells, the baseline gaining more, and the paired Δgap **varies in sign** by corpus — so "the convention understates the effect" is not claimed. On five arm-cells at n = 10 it is realized on a disjoint split at **23.3 %** (claim-level **[−17.3 %, 53.2 %]**, P(rate ≤ 0) = **0.118**). Two things do **not** hold: the σ√(2 ln E) scaling was tested and **withdrawn**, and no power analysis was registered, so the nine `n = 3` cells stay descriptive. What we ask of a detection paper is narrow: **name the counting unit, freeze in writing which checkpoint is reported, and state the interval at the claim's level.** First report is the companion's (§7.2).

## Declarations

- **Data availability.** The **seven experiment corpora are open**; what the pack lacks is their **images**, so retraining is not recomputable (§10 defect 6). Separately, §5.5's **third-party logs** are **not redistributed** (hashes + pointers only), as **5 of their 6 sources state no licence**. The package — data, per-epoch matrix, verdicts, checkers, `MANIFEST_sha256.csv` — is at **https://github.com/YBP2005/P2.git** (accessed 2026-09-29). The reproduction package itself — data files, per-epoch matrix, verdict files, checkers, with their SHA-256 list in `MANIFEST_sha256.csv` — is deposited and openly available at **https://github.com/YBP2005/P2.git** (accessed 2026-09-29).
- **Recomputability boundary.** Every reported number is recomputable from the released per-run artifacts **without retraining**, each with its script; **three are not, and they are named**: the corpora and carve directories the archive **holds no copy of** (open corpora, images not in the pack; §10 defect 6), the conflicting-endpoint archives (§7.3), and the `aitod20` completion to n = 10.
- **Competing interests.** None declared.
- **One quantity, two spellings.** The n = 10 attainable floor is **exactly $2/2^{10}$ = 0.001953**; the tables use the exact value, the abstract writes **0.002**. Same floor, spelling only.
- **Reproducibility.** Every number points to a local artifact (**artifact paths in this article and in the Supplementary are relative to the released `02_release_data/` directory**); **289 audit anchors** mechanically guard the claims, and six checkers (`audit_anchors_fupaper.py`, `check_crosscite.py`, `verify_pulled.py`, `gap_mechanism_20260916.py`, `verify_pr_forms_20260917.py`,

`verify_submission_pack_20260917.py`) ship with the archive: **one runs unaided for a holder of the archive; four are author-side gates; and the remaining one -- the anchor audit itself -- is partly runnable, checking 270 of the 289 anchors from the released copy alone (the other 19 need the authors working-tree inputs)**. The archive is the one deposited above.

## References

> **Note on preprint citations.** Seventeen of the 55 entries give an arXiv edition; **three** also name a peer-reviewed venue and **fourteen** have none. Those fourteen cover four benchmark or dataset canonical releases, two methodological works and eight concurrent or tooling works; **all are cited for positioning only**, and every number comes from our own corpora or the third-party logs of §5.5. The full enumeration is in the supplement.

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